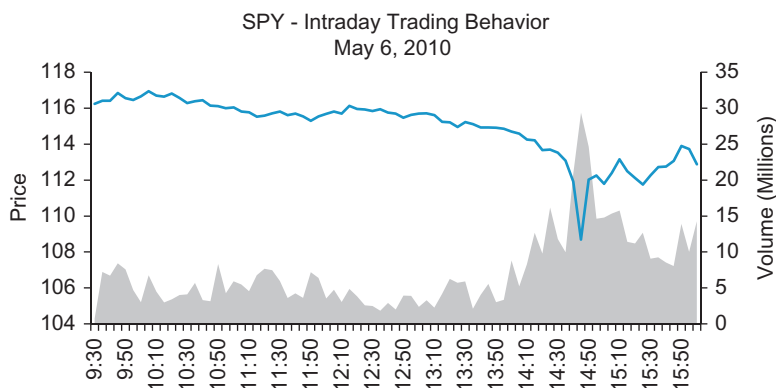




■ **Figure 2.8** SPY Intraday Trading Behavior.

approximately 1 p.m. prices started to decline slowly. Around 2:40 p.m. there was a very sharp decline in prices that lasted for about 20 minutes, followed by a price reversal. SPY did experience another slight decrease towards the close but this was due to offsetting positions that were acquired during the day. Volume in SPY followed almost an identical pattern. Volumes were pretty constant until about 2:00 p.m. when they started to increase. Then there was the sharp increase in volume traded at the time of the flash crash right around 2:40 p.m. Volumes leveled off slightly after the reversal but still remained much higher than normal into the close.

Our sample of ten stocks has some similar patterns to SPY (Table 2.12). The average volume ratio was 3.6 compared to 3.3 (SPY). The close-to-close price change was -3.4% (same as SPY) and the open-close change was -4.3% compared to -2.9% for SPY. The most dramatic difference, however, was the magnitude of the high-low price change for these stocks. The average percentage high-low price change was -35.3% which was 3.3 times higher than the high-low percentage range for SPY.

The intraday price movement (normalized starting at 100) for these names is shown in Figure 2.9. Notice that most of the stocks had price movement that was relatively constant up until about 2:40 p.m. Some stocks opened much lower or declined over the first half-hour but these also remained pretty constant through the flash crash. Each experienced a very sudden and sharp decline which was quickly followed by a reversal. Note how quickly prices declined for all of these stocks only to be followed by another quick and sharp reversal. The close-to-close price change of -3.4% was equivalent to a -2.1 sigma event. On a 250 day trading year we should expect declines of